

The background features decorative geometric shapes in the corners. In the top-right corner, there is a solid dark blue circle and a circle with dark blue diagonal stripes. In the bottom-left corner, there is a solid dark blue circle and a circle with dark blue diagonal stripes. In the bottom-right corner, there is a solid dark blue circle.

Research Project

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Research Question

The study seeks to determine how distance from the Melbourne CBD affects median weekly rental prices within the city.

- Economic relevance: Proximity to CBD influences residential choices, guiding urban planning and infrastructure development around housing clusters.
- Historical significance: CBD was once the main employment hub, creating a steep negative rent-distance gradient.



Data

Variables:

- Distance (explanatory): straight-line distance (km) from each LGA's centroid to Melbourne CBD.
- Median rent (response): the LGA-level mean of annual median rents (AUD/week).
- Crime (control): mean annual offence count per LGA.
- Schools (control): total school count within each LGA.
- Health (control): number of GP clinics per 1,000 residents in LGA.

Sources: Government Department Data + AI generated straight-line distance

Data cleaning: 79 → 31 LGAs

- Exclusion of Rural LGAs & maintaining High-Leverage LGAs
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Empirical strategy

$$\overline{median_rent}_i = \beta_0 + \beta_1 \overline{distance}_i + \beta_2 \overline{crime}_i + \beta_3 \overline{schools}_i + \beta_4 \overline{health}_i + \varepsilon_i$$

Descriptive analysis:

estimate the direction, strength, and consistency of the association between distance and rent

Assumptions:

1. LGAs properly capture Melbourne's spatial structure
2. Controls include the most relevant amenities (crime, schools, health) to reduce OVB
3. Linear form is reasonable – tested with log-rent robustness check

Result

Table 1. Main Regression Results (HC3 Standard Errors)

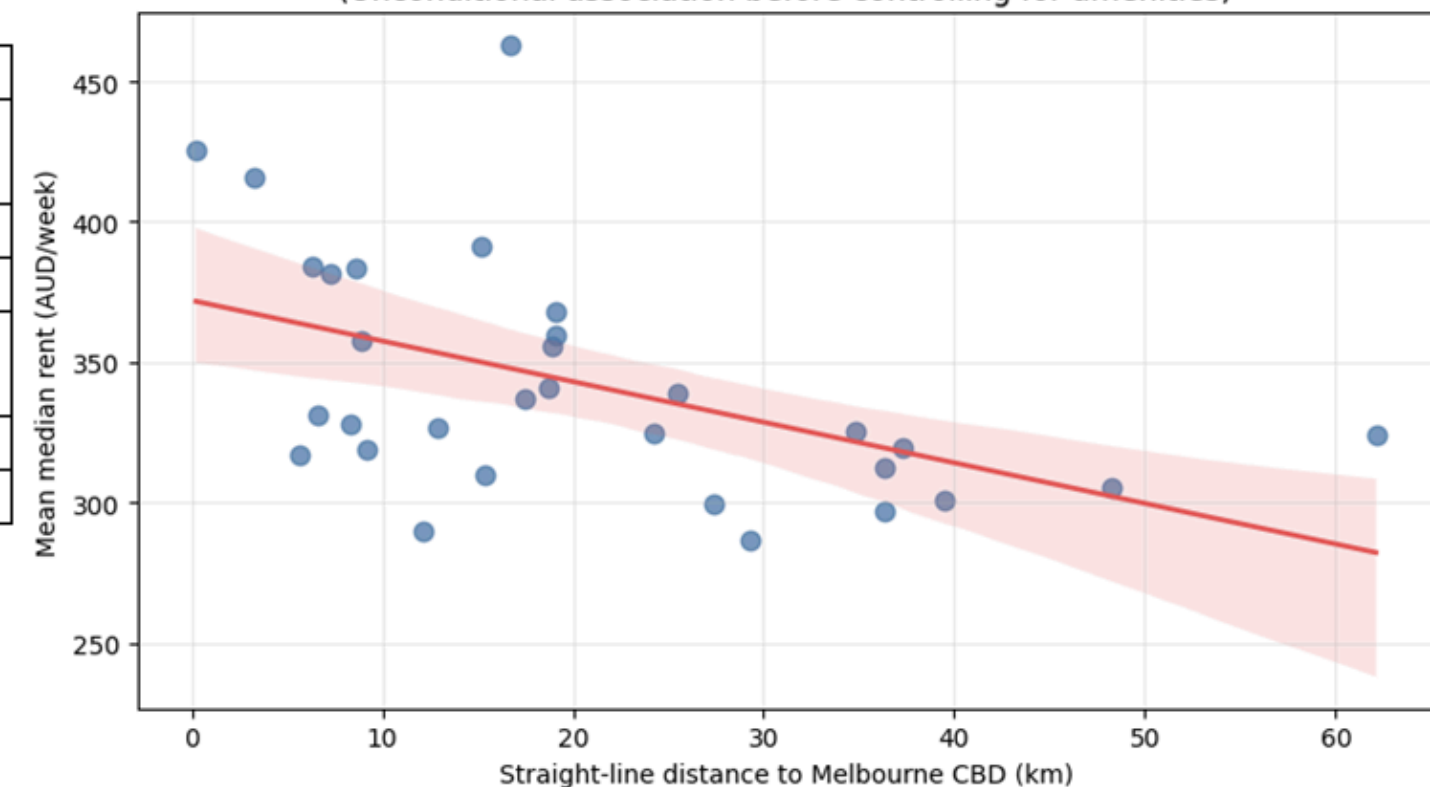
Dependent Variable: Median Weekly Rent (AUD/week), N=31 LGAs

Variable	Coefficient	Standard Error	p-value
Distance (straight-line km)	-0.8174	0.567	0.161
Crime (offence count)	-0.0011	0.001	0.419
Schools (count)	-0.3709	0.541	0.499
Health access (GP clinics per 1000)	91.8932	77.940	0.249
Constant	359.6227	37.625	<0.001
R^2	0.355		

Interpretation of results:

- Distance coefficient: -0.8174
- Economic implication (e.g. for 20km away from CBD): ≈\$16.40 p/w, ≈\$850 p/year
- However, $R^2 = 0.355$ (35.5% of variation in rent explained by variation in distance)
- Mean Rent vs Distance from CBD Scatterplot (High Dispersion)
- Weak to Moderate goodness of fit
- Reason: other variables

Figure 1: Mean Rent vs. Distance from CBD
(Unconditional association before controlling for amenities)



Robustness

Table 2: Condensed Robustness Table

Specification	Distance (β)	Standard error	p-value	Interpretation
(1) Minimal Distance only	-1.4461	0.5344	0.0068	Strong negative gradients without controls
(2) Preferred crime + schools + health	-0.8174	0.5279	0.1215	Negative but imprecise, controls attenuate affect
(3) Kitchen-sink controls (n obs reduced)	-0.3696	0.5269	0.4830	Further attenuation with additional amenities
(4) Drop high-leverage ude 4 high-leverage LGAs	-1.2861	0.5339	0.0160	Sign remains negative; significance increases
(5) Log rent % change in rent per km	-0.0025	0.0016	0.1237	-0.25% per km, same direction
(6) Permutation placebo 1000 random permutations	-0.8174	0.4955	0.0950	Observed effect unlikely under random assignment.

Key takeaway: Negative sign never flips → directionally robust



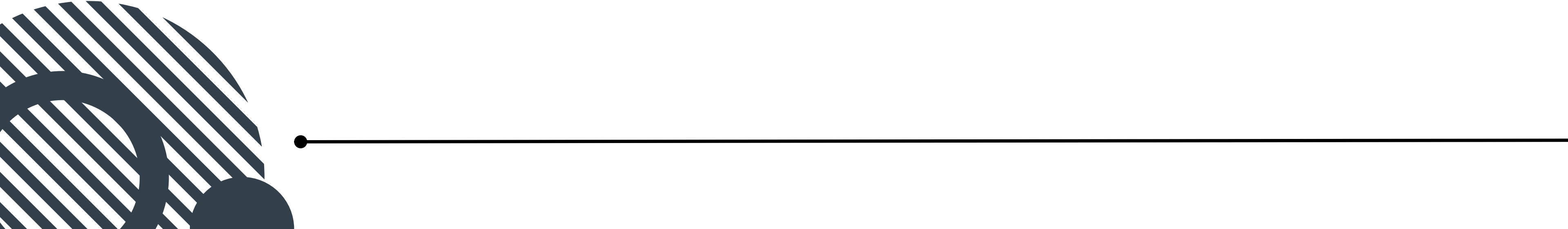
Limitations and Conclusion

1. LGA level data

- Benefit: easy comparison, meaningful size (total schools, offence count etc), available data
- Issue: OVB (not all variables have available LGA data e.g. transport accessibility)

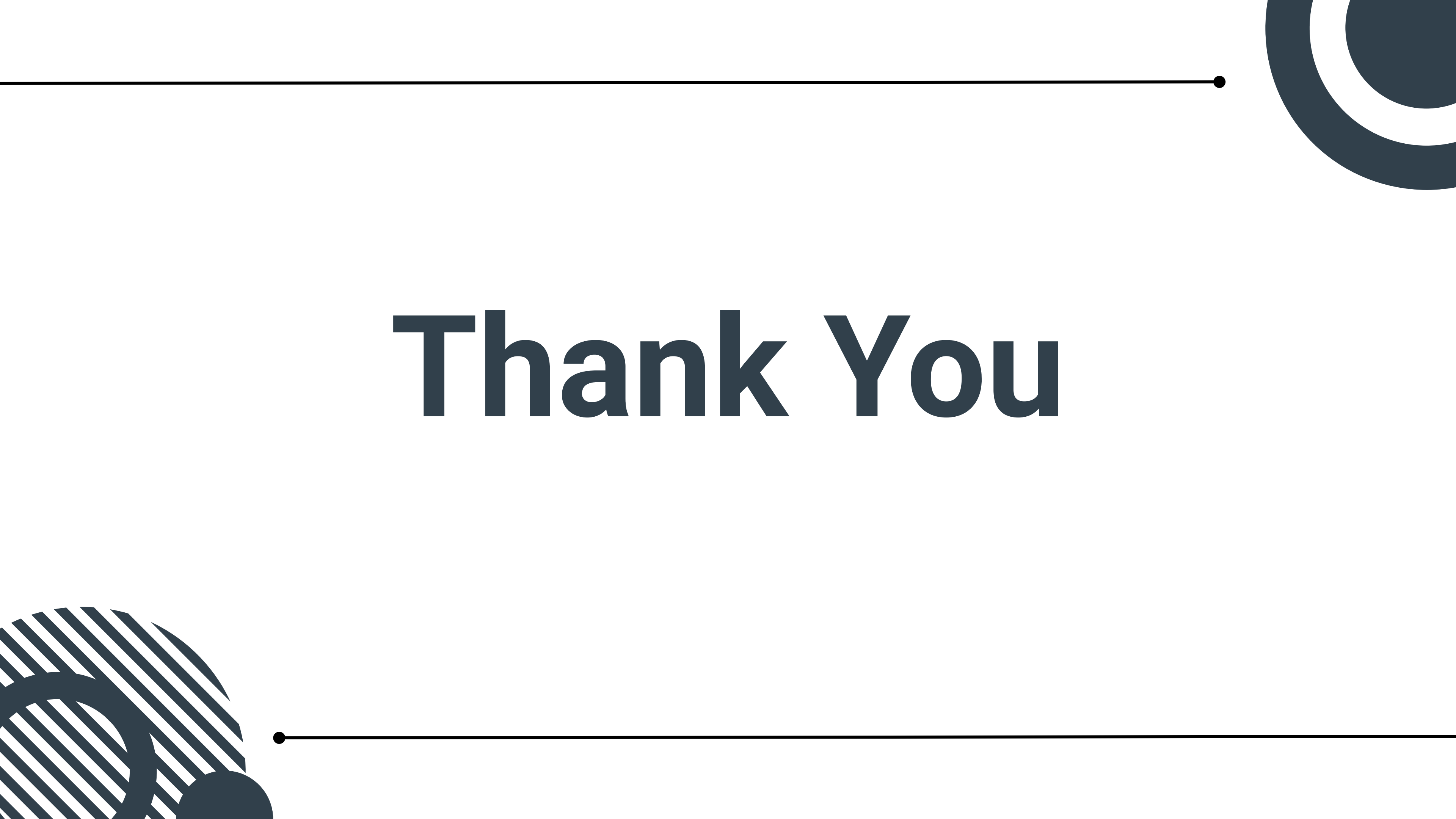
2. Straight-line distance may not reflect real commuting accessibility.

Future solutions:

- Access higher quality spacial data or individually collect stronger data
 - Reduces OVB, creating stronger controls.
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Reference

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Thank You
